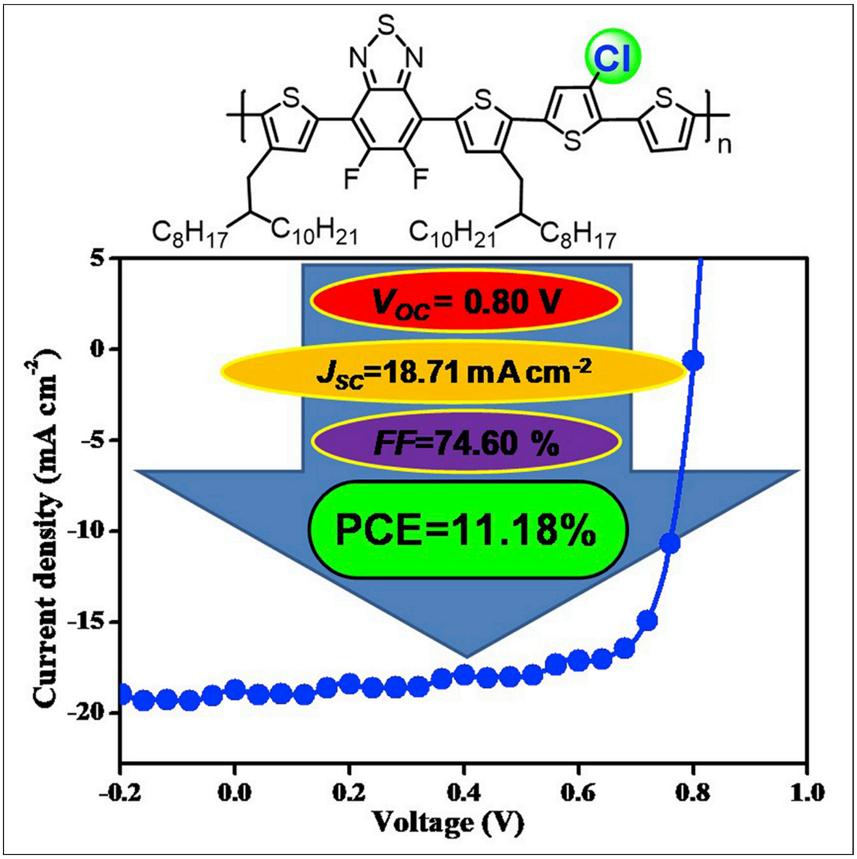
Joule

Article

A Chlorinated π-Conjugated Polymer Donor for Efficient Organic Solar Cells



We reported a chlorinated polymer donor **PBT4T-Cl** that showed a high PCE of 11.18%. The results demonstrated chlorination of the thiophene moiety could fine-tune the HOMO level of the corresponding polymer as well as the carrier mobility and morphologies of the blend film, eventually improving its photovoltaic properties as a donor without thermal annealing.

Hui Chen, Zhiming Hu, Huan Wang, ..., Wei Chen, Anhua Liu, Feng He

hef@sustc.edu.cn

HIGHLIGHTS

A chlorinated polymer **PBT4T-Cl** is developed with simple and enlargeable synthesis

The chlorination leads to enhancements both in V_{OC} and FF, with a PCE up to 11.18%

The chlorinated polymer-based PSCs exhibit favorable stability in lifetime tests

Chen et al., Joule 2, 1623–1634
August 15, 2018
https://doi.org/10.1016/j.joule.2018.05.010

Joule

CellPress

Article

A Chlorinated π-Conjugated Polymer Donor for Efficient Organic Solar Cells

Hui Chen,[1,5] Zhiming Hu,[1,4,5] Huan Wang,[1] Longzhu Liu,[1] Pengjie Chao,[1] Jianfei Qu,[1] Wei Chen,[2,3] Anhua Liu,[4] and Feng He[1,6,*]

SUMMARY

We synthesized a chlorinated benzothiadiazole-T4 polymer donor PBT4T-Cl, in which a chlorine atom had been introduced at the 4 position in the middle thiophene unit to fine-tune the energy level of the final polymers. Compared with its non-chlorinated analog, the PBT4T-Cl-based devices exhibited clear increases in open-circuit voltage and fill factor, achieving power conversion efficiencies (PCEs) up to 11.18% with simple synthesis, which is the highest PCE of the chlorinated polymer-based fullerene polymer solar cells (PSCs) reported to date. Grazing incident wide-angle X-ray scattering, atomic force microscopy, and transmission electron microscopy measurements revealed an optimized morphology of the spin-coated PBT4T-Cl blend films, which supported that chlorine substitution could promote the performance of PSCs. More importantly, the PBT4T-Cl-based devices showed superior stability, with a PCE of 8.16% after 50 days' device storage. Through this research, the chlorination of low-band-gap polymers provides new insight into designing π-conjugated polymer semiconductors and realizing further enhancement of polymer solar cell efficiency as well as stability.

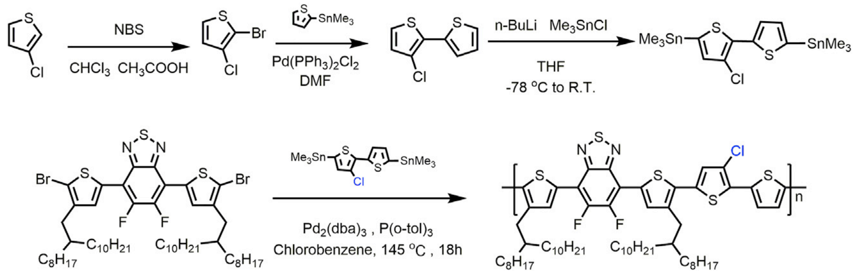
Context & Scale

Polymer solar cells (PSCs) could be fabricated on a large scale using solution processing at low cost, and thus are a promising energy source that can be exploited globally to contribute to worldwide electricity needs. Fluorination was applied successfully to PSCs, attributed to an enhanced V_{OC}. However, fluorination has some defects, including lower reaction yield and the use of hazardous fluorination reagents. To balance the dilemma, chlorination has been explored in organic photovoltaics because chlorine is another electronegative halogen with much easier chemical synthesis and low raw-material cost. In this study, we developed a new chlorinated benzothidiazle-T4 polymer **PBT4T-Cl** with a high power conversion efficiency of over 11% and enhanced shelf stability benefiting from its improved intermolecular interactions through the halogen bonds. This research contributes a useful chlorination strategy for regulating the desirable photovoltaic properties of polymers.

INTRODUCTION

Polymer solar cells (PSCs) based on a blend of a semiconducting conjugated polymer donor and an electron acceptor have attracted enormous attention as solution-processable, lightweight, flexible, and inexpensive energy sources.[1–11] With the development of material engineering, interface modification, and advanced device processing in past decades, the power conversion efficiency (PCE) of the state-of-the-art PSCs has already exceeded 14%.[12–15] Notably, the development of low-band-gap donor-acceptor (D-A)-type polymers boosts the PCE enhancement of bulk heterojunction (BHJ) PSCs.[15–19] To achieve outstanding device performance, the intrinsic properties of donor polymers, including their band gap (*Eg*), frontier energy levels, and solubility, must be carefully considered.[20] Narrow band gaps and deep HOMO (highest occupied molecular orbital) levels are the most common and practical strategies for polymer donor design, being aimed at effectively harvesting solar photons and attaining high open-circuit voltages (V_{OC}).[20–22] At the same time, neat crystalline structures and the desired backbone orientation of the polymer are also essential to achieve an optimal morphology of the active layer, relating to efficient dissociation of excitons and carrier transportation.[23–26]

Among the many efficient donor polymers, the D-A conjugated polymers with fluorinated benzothiadiazoles (FBTs) have been extensively investigated and demonstrate great potential as superior polymer donor materials.[5,27–30] To the best of



Scheme 1. The Key Synthesis Routes for PBT4T-Cl

3-Chlorothiophene was bromized by NBS and then coupled with a thiophene through Stille coupling reaction to give the product 3-chloro-2,2′-bithiophene. DMF, dimethylformamide; R.T., room temperature; THF, tetrahydrofuran.

our knowledge, the most impressive FBT-based polymer is PffBT4T-2OD, reported by Yan's group. The thick-film PSCs based on PffBT4T-2OD and traditional $PC_{71}BM$ had a high PCE above 10%; the aggregation and morphology control of their blend films was critical to the performance of their PSCs[5]; and the family polymers of PffBT4T have been widely study in fullerene and non-fullerene PSCs,[10,31,32] achieving many advances. Although the PffBT4T-2OD devices possessed excellent PCEs with very high short-circuit currents (J_{sc}) and fill factors (FFs), the moderate V_{OC} limited the pursuit of further improvements of the PCE. This is based on the origin of V_{OC} in PSCs, which is determined by the HOMO level (E_H) and photon energy loss ($E_{loss} = E_g - eV_{OC}$).[33,34] Motivated by this, the newly designed PffBT4T-2OD-related polymer would have a deeper E_H and a lesser E_{loss} in PSCs. Incorporating fluorine into conjugated polymers is a practical method to improve the V_{OC} of PSCs because of the lower HOMO levels and more highly oriented film structures thus attainable compared with those of their non-fluorinated counterparts.[35–37] Fluorination has been successfully used in multifarious organic semiconductors, including polymers and small molecules, in the past decade. Compared with fluorine, chlorine is another electronegative halogen with much easier chemical synthesis and low raw-material cost. Chlorinated materials have been studied in organic field-effect transistors (OFETs)[38–40] and organic photovoltaics[13,15,41–43] because of their appropriate energy levels and suitable mobilities. The performances of chlorinated polymer-based PSCs have been beyond those of fluorinated materials in some special polymer systems.[13]

In this study, a chlorinated FBT-based polymer was designed and synthesized by incorporating a chlorine atom at the 4 position of the middle thiophene unit. The chemical structure of this compound, named **PBT4T-Cl**, is shown in Scheme 1. As a polymer donor, inverted fullerene (6,6-phenyl C_{71} butyric acid methyl-ester [$PC_{71}BM$]) PSCs with 2% 1,8-diiodooctane (DIO) additive showed a high PCE of 11.18% with an elevated open-circuit voltage of 0.80 V, which is the highest efficiency reported to date of a chlorinated polymer-based fullerene PSC and is also one of the highest PCEs among the reported FBT-based PSCs. Even without the additive, the new chlorinated polymer attained a PCE of 10.17%, which is approximately 34% higher than that of the non-chlorine control device without any additive and is also slightly higher than that of the optimized non-chlorine control device with 3% additive from our parallel device testing. More importantly, the **PBT4T-Cl**-based devices showed superior stability compared with the non-chlorine analog. **PBT4T-Cl**-based devices showed almost no change of V_{OC} during 50 days of testing; the PCE of **PBT4T-Cl** remained at approximately 8.16% after the 50 days of storage. The effects of the chlorine substitution on the optical properties, morphology of

[1]Department of Chemistry and Shenzhen Grubbs Institute, Southern University of Science and Technology, Shenzhen 518055, P.R. China

[2]Materials Science Division, Argonne National Laboratory, 9700 Cass Avenue, Lemont, IL 60439, USA

[3]Institute for Molecular Engineering, The University of Chicago, 5640 South Ellis Avenue, Chicago, IL 60637, USA

[4]Advanced Materials Laboratory, Key Laboratory of High Performance Ceramic Fibers of Ministry of Education, College of Materials, Xiamen University, Xiamen 361005, P.R. China

[5]These authors contributed equally

[6]Lead Contact

*Correspondence: hef@sustc.edu.cn

https://doi.org/10.1016/j.joule.2018.05.010

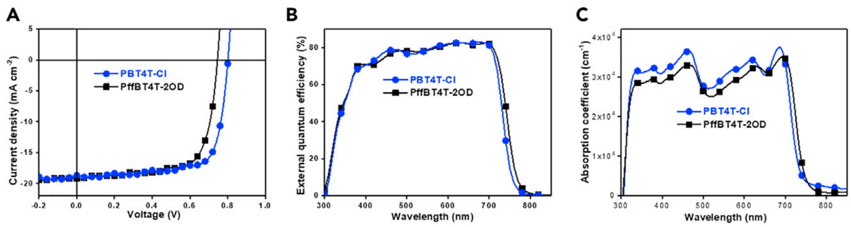


Figure 1. The Steady-State Photovoltaic Properties of Polymers Blended with $PC_{71}BM$

(A) *J-V* curves of the best PSC devices based on **PBT4T-Cl** and PffBT4T-2OD blended with $PC_{71}BM$ under 100 mW cm^{-2} AM 1.5 G irradiation.

(B) The EQE spectra of corresponding devices.

(C) The absorption coefficient spectra of corresponding active layers.

blend films, and charge dynamics were carefully studied to understand the role of chlorination in high-performance PSCs. Collectively, the above results indicate that chlorination is a promising route to reduce the material cost with simple and enlargeable synthesis, and to V_{OC} enhancement for better device performance, especially with suitable polymer backbone bonding.

RESULTS AND DISCUSSION

Photovoltaic Performance

To evaluate the potential of chloride **PBT4T-Cl**, it was synthesized by Stille coupling polymerization at gram scale in the laboratory with an average number molecular weight (M_n) of 23.7 kDa and a polydispersity index (PDI) of 1.39. The key synthesis routes are shown in Scheme 1. 3-Chlorothiophene was bromized by N-bromosuccinimide (NBS) and then coupled with a thiophene through Stille coupling reaction to give the product 3-chloro-2,2′-bithiophene. Later this product underwent stannylation and was coupled another monomer through Stille coupling reaction in chlorobenzene to give the polymer. PffBT4T-2OD was synthesized with an M_n of 46.7 kDa and a PDI of 1.51. **PBT4T-Cl** showed good thermal stability with a decomposition temperature (T_d, 5% weight loss) at 445°C from thermogravimetric analysis (TGA) measurements (See Figure S1), which is much higher than that of its non-chlorine analog PffBT4T-2OD (412°C). The PffBT4T-2OD-based devices were fabricated following a previous published report.[5] The literature demonstrates that PffBT4T-2OD-based PSCs are high efficiency (>10%), high FF (>70%), and thick film (~300 nm). From the parallel experiments of our work, the chlorinated **PBT4T-Cl** inherits these outstanding properties and further realizes higher PCE levels in its PSCs.

The characteristic current density-voltage (*J-V*) curves and the external quantum efficiency (EQE) for the best devices are shown in Figures 1A and 1B, respectively. The extracted parameters of performance for **PBT4T-Cl** with different thickness and optimized PffBT4T-2OD devices are listed in Table 1. As expected, the device based on **PBT4T-Cl** and $PC_{71}BM$ exhibited a higher V_{OC} of 0.80 V compared with that of PffBT4T-2OD (0.75 V)-based PSCs due to its lower HOMO energy level. The **PBT4T-Cl**-based optimal device with thickness of 280 nm showed the higher PCE of 11.18%, with a J_{sc} of 18.71 mA cm^{-2} and a higher FF of 74.60% without thermal annealing. The control PffBT4T-2OD also showed a decent PCE of 10.17%, with a higher J_{sc} of 19.23 mA cm^{-2} and an FF of 70.54%, approaching the optimal reported performance.[5] Therefore, the enhancement of the photovoltaic performance of **PBT4T-Cl** demonstrates that the chlorinated thiophene moiety is significant for polymer donor design, being facile and practical for improving V_{OC} and FF while

Table 1. The Performance Parameters of PSC Devices Based on PBT4T-Cl with Different Thickness and PffBT4T-2OD with Optimal Thickness under 100 mW cm^{-2} AM 1.5 G Irradiation

Polymers	Thickness (nm)	V_{OC} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
PBT4T-Cl	165	0.80 (0.80 ± 0.003)	16.57 (16.45 ± 0.17)	66.85 (65.86 ± 0.88)	8.86 (8.72 ± 0.17)[b]
	220	0.80 (0.80 ± 0.004)	17.53 (17.31 ± 0.21)	72.71 (71.51 ± 1.06)	10.19 (9.98 ± 0.21)[b]
	280	0.80 (0.80 ± 0.005)	18.71 (18.14)[a] (18.42 ± 0.29)	74.60 (72.31 ± 1.96)	11.18 (10.92 ± 0.28)[b]
	320	0.80 (0.80 ± 0.005)	18.97 (18.62 ± 0.34)	71.64 (70.29 ± 1.77)	10.82 (10.59 ± 0.32)[b]
	390	0.81 (0.80 ± 0.006)	19.21 (18.82 ± 0.41)	65.65 (64.31 ± 1.18)	10.21 (10.02 ± 0.27)[b]
PffBT4T-2OD	290	0.75 (0.75 ± 0.005)	19.23 (18.76)[a] (18.91 ± 0.36)	70.54 (69.11 ± 1.38)	10.17 (9.87 ± 0.28)[b]

[a]The calculated J_{sc} values from EQE curves.
[b]Average value ± SD from the statistics of 30 different devices.

sacrificing only a slight amount of J_{sc}. To fabricate the devices on a large scale, the device performance needs to be insensitive to the active layer thickness.[28,44] Thus, the relationship to photovoltaic performance and thickness of blend films was studied. As presented in Table 1, the value of FF increased from 66.85% at 165 nm to 74.60% at 280 nm. The dominant factor is strong temperature-dependent aggregation of polymer with 2-octyldodecyl alkyl chains in quaterthiophene, and the thicker active layer has high-quality film morphology during the spin-coating process.[5] As expected, the **PBT4T-Cl**-based devices have high efficiency (>10.19%) for thicknesses ranging from 220 nm to 390 nm (see Table 1 and Figure S2), which is suitable for roll-to-roll device fabrication.

The EQE curves of the polymer-based devices show a tendency consistent with the observed variation in J_{sc}. The **PBT4T-Cl**-based device has a slightly narrower range of spectral response than the PffBT4T-2OD device does, as shown in Figure 1B, attributed to the slight blue shift of its absorption coefficient spectra (Figure 1C). Nonetheless, the EQE values of all devices are comparable between 300 and 700 nm, especially where they are very high (~80%) from 500 to 700 nm. The calculated integrating current from EQE curves matches well with those of the *J-V* tests. The very small change of J_{sc} indicates that the substitution of chlorine atoms could not notably hinder the efficient exciton dissociation and charge transport.

The origin of the increased V_{OC} and decreased J_{sc} of **PBT4T-Cl** was investigated by UV-visible absorption spectra. Compared with the PffBT4T-2OD film, the chlorination led to obvious changes in the optical properties in thin films as casted. As seen in Figure S3, both polymers displayed two absorption bands: the weak band I at 300–500 nm, and the intensive band II at 500–800 nm. It is clear that both polymers are relatively low-band-gap donors for highly efficient PSCs. The two polymers have much the same features in band I, indicating that the chlorination barely alters the π-π^* transition of the conjugated backbone.[33] Meanwhile, **PBT4T-Cl** exhibited a slight blue shift by 8 nm at λ_{max} (692 nm) in comparison with PffBT4T-2OD in band II, which resulted in a marginally narrower absorption spectrum of **PBT4T-Cl** and produced less efficient light harvesting and slightly decreased J_{sc} in the final PSCs. The blue shift perturbation should arise from the lessened coplanarity between the chlorinated thiophene and the FBT units due to the increased steric hindrance from the larger atomic radius of chlorine. Consequently, the intermolecular π-π stacking interactions of **PBT4T-Cl** should also have lower intensity than those of PffBT4T-2OD, resulting in the deep HOMO level of the polymer and enhanced V_{OC} of corresponding devices.[44] In addition, from the onset of absorption, the optical band gaps of the films were estimated to be 1.67 eV for **PBT4T-Cl** and 1.65 eV for PffBT4T-2OD, as listed in Table S1. The amplification of enhanced V_{OC}

Table 2. The Calculated Dipole Moments for Repeating Units of PBT4T-Cl and PffBT4T-2OD

Polymers	μ_g (D)	μ_e (D)	$\Delta\mu_{ge}$ (D)
PBT4T-Cl	3.12	11.61	10.78
PffBT4T-2OD	2.92	10.35	9.55

μ_g, ground-state dipole moment; μ_e, excited-state dipole moment; $\Delta\mu_{ge}$, difference in ground-state and excited-state dipole moment.

was larger than the change range of band gap. Thus, the **PBT4T-Cl** has a lower E_{loss} than PffBT4T-2OD does (see Table S2), demonstrating that polymer chlorination reinforces the effective conversion of absorbed photons. To further support this point, the energy levels of the polymers were examined by cyclic voltammetry. As shown in Table S3, and Figures S4 and S5, the tested results accord with the blue shift of the absorption onset caused by the polymer chlorination. To show the experimental results and have a better understanding of the relationship between structure and properties, the optimized geometries (Figure S6) and molecular frontier orbitals (Figure S7) for polymers were calculated by density functional theory (DFT) simulations with the Gaussian 09 program at b3lyp/6-311G* level.[45–48] The calculations indicate that chlorination lowers HOMO energy level and hardly changes the planarity of backbone geometries, but the chlorine is also an electrophilic candidate with large atomic radius, so the dipole moments could be strong influenced. The $\Delta\mu_{ge}$ was also calculated by DFT following the previous reports,[45,47] and is presented in Table 2. The dimer of **PBT4T-Cl** shows a larger $\Delta\mu_{ge}$ value (10.78 D) than that of PffBT4T-2OD (9.55 D), and the higher dipole moment reduces the Coulomb binding of excitons. As discussed later, the probability of exciton dissociation is elevated and mobility of charge carriers is improved.[48]

The space-charge-limited current method[49] was used to evaluate the hole and electron mobility of the active layers in the PSC devices (see Figure S8 and Table S4). The **PBT4T-Cl** blend film with 280 nm achieved a high hole and electron mobility of 2.6×10^{-3} cm^2 V^{-1} S^{-1} and 1.8×10^{-3} cm^2 V^{-1} S^{-1} by the addition of the chlorine atoms, while the PffBT4T-2OD blend film with 290 nm also showed a fair hole and electron mobility of 1.1×10^{-3} cm^2 V^{-1} S^{-1} and 5.9×10^{-4} cm^2 V^{-1} S^{-1}. Improved mobility and charge balance produces favorable charge transport and efficient carrier collection, resulting in improvements in FF.[50]

Morphology Study

Recently, thin film morphology factors, including domain size of phase separation, molecular ordering stacking, the degree of aggregation, crystallinity, and the miscibility or distribution of donor/acceptor within photoactive layer, have been shown to be crucial factors in determining the photovoltaic performance in PSCs.[25,51] To further shed light on the role of inserting chlorine atoms into thiophene moiety-based semiconducting polymers from the perspective of ordering structure and film morphology, we present detailed investigations on the morphology of blend films by grazing incident wide-angle X-ray scattering (GIWAXS), atomic force microscopy (AFM), and transmission electron microscopy (TEM).

To explore the structure-performance relationship in the active layer of the films, the microstructure of the **PBT4T-Cl** and PffBT4T-2OD blend with $PC_{71}BM$ was examined by GIWAXS (see Figures 2A and 2B); the patterns of the control PffBT4T-2OD is in agreement with the referenced literature.[5] For the chlorinated blend films of **PBT4T-Cl**:$PC_{71}BM$, the crystalline content was found to be further enhanced, because the calculated full width at half-maximum (FWHM) of

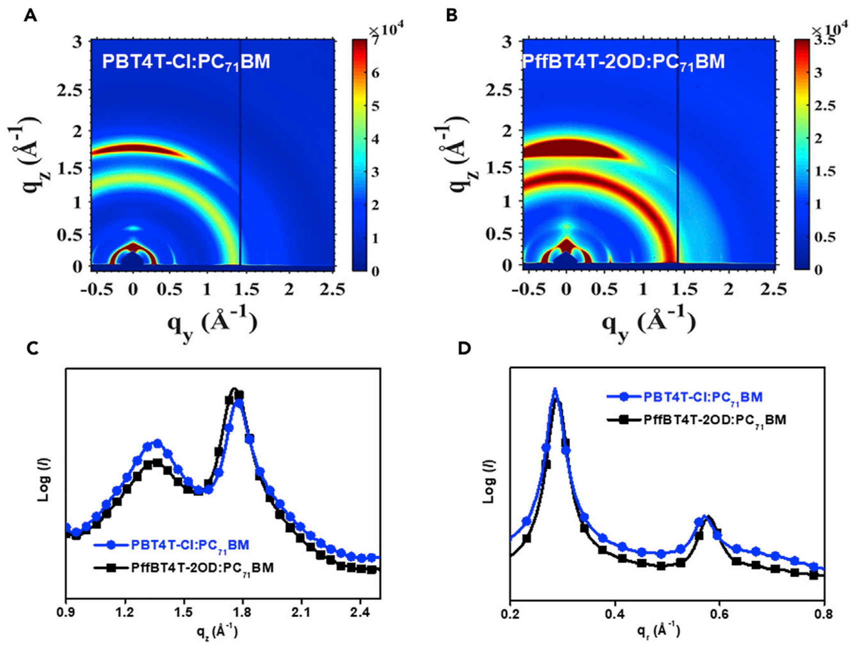


Figure 2. The GIWAXS Characterization of Polymer Blend Films
(A) GIWAXS patterns of **PBT4T-Cl**:$PC_{71}BM$ blend film.
(B) GIWAXS patterns of PffBT4T-2OD:$PC_{71}BM$ blend film.
(C and D) Out-of-plane (C) and in-plane (D) cut profiles of **PBT4T-Cl**:$PC_{71}BM$ and PffBT4T-2OD:$PC_{71}BM$ blend films, respectively.

PBT4T-Cl is 0.12 $Å^{-1}$ at a scattering peak of ~1.76 $Å^{-1}$, the FWHM of PffBT4T-2OD is 0.19 $Å^{-1}$. Notably, a sharper arc-like peak was observed in the out-of-plane direction (q_z), corresponding to an almost identical π-π stacking distance of 3.51 Å in **PBT4T-Cl**:$PC_{71}BM$ blending film (see Figure 2C). As implied by the (100) and (010) diffraction in the q_z and q_r, the π-π stacking direction of the polymer in the blend film appeared to produce a more prominent face-on orientation. Interestingly, the chlorinated thiophene in the backbone of polymer has no negative effects in molecular orientation. A possible factor could be that the chlorine atoms bonded with the thiophene units, even with the larger atomic radius of the chlorine, will benefit the formation of strong lamellar stacking in the microstructure of the blend films. In addition, a slightly steeper scattering peak of **PBT4T-Cl** was shown in the in-plane direction (q_r) at ~0.29 $Å^{-1}$ (see Figure 2D), indicating the larger crystallite size of **PBT4T-Cl** in the blend film according to a narrower FWHM of **PBT4T-Cl** compared with that of PffBT4T-2OD ($FWHM_{PBT4T-Cl}$ = 0.031 $Å^{-1}$, $FWHM_{PffBT4T-2OD}$ = 0.047 $Å^{-1}$, respectively).[52] To support the above viewpoint, we also tested GIWAXS patterns of pristine polymer films, with the similar behaviors observed in Figure S9. On the other hand, **PBT4T-Cl** exhibited a stronger aggregation than PffBT4T-2OD from the different temperature absorption spectra[5] (see Figure S10). The peak of **PBT4T-Cl** located at 692 nm was caused by the intensive aggregation, and the interchain $\pi-\pi^*$ transition peak reduced gradually and finally disappeared as the temperature elevated from 30°C to 110°C, but the peak of PffBT4T-2OD disappeared at 70°C. Thus, the incorporation of chlorine into the polymer provided a practical method to enhance the intermolecular π-π interactions in the active layer film. This helped achieve well-defined orientation both morphologically and in the crystalline domains of the active layer at the nanoscale, which is favorable for efficient diffusion of excitons to the donor-acceptor

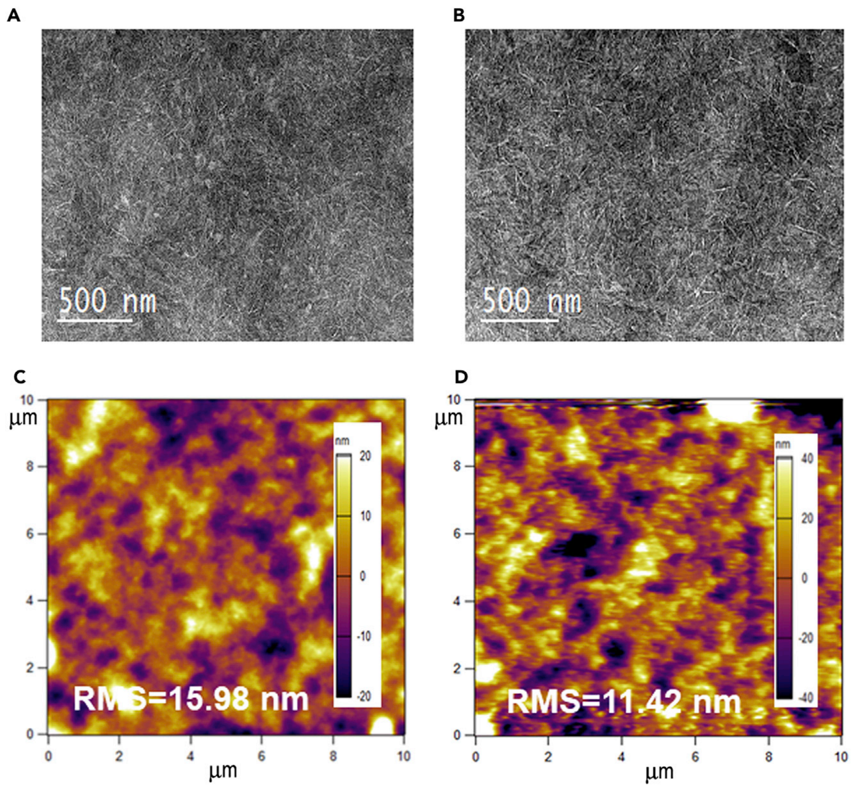


Figure 3. The Nanostructural Morphology of Polymer Blend Films

(A and B) TEM images of **PBT4T-Cl**:$PC_{71}BM$ (A) and PffBT4T-2OD:$PC_{71}BM$ blend film (B) with DIO additive processing, respectively. Scale bars: 500 nm.

(C and D) AFM images of **PBT4T-Cl**:$PC_{71}BM$ (C) and PffBT4T-2OD:$PC_{71}BM$ (D) blend films.

interface and the concomitant exciton dissociation, leading to favorable charge transport and high FF, as discussed above.[5,25]

Moreover, TEM was used to research the extent and length of the phase separation (Figures 3A and 3B). Interestingly, the as-cast film of the **PBT4T-Cl**:$PC_{71}BM$ blend using DIO as the additive directly displayed a similar uniform and favorable fiber nanostructure (Figure 3A) compared with that of PffBT4T-2OD:$PC_{71}BM$ after annealing (Figure 3B), which is good for efficient exciton diffusion. Both the **PBT4T-Cl** (as cast) and PffBT4T-2OD (after annealing) blend films without DIO processing exhibited just passable fiber nanostructure (Figures S11A and S11B). These interesting results reveal that the chlorine-substituted polymer in **PBT4T-Cl** could induce the desired morphology with enlarged interfacial areas and bicontinuous interpenetration without thermal annealing because of the synergistic effect between the large spacer effect and the high electronegativity of the chlorine atoms. Furthermore, AFM with tapping mode was used to study the surface morphologies of the blend films with or without DIO processing (Figures 3C, 3D, S11C, and S11D). For the blend films without DIO treatment, the root-mean-square (RMS) roughness values were 8.06 and 4.27 nm, corresponding to **PBT4T-Cl** and PffBT4T-2OD, respectively. After treatment by DIO processing, the RMS value of PffBT4T-2OD:$PC_{71}BM$ blend film was up to 11.42 nm (Figure 3D) after the annealing treatment. The rougher surface is favorable for efficient diffusion of excitons to the donor-acceptor interface and the concomitant exciton dissociation and resulted in the PCE of PffBT4T-2OD-based PSC increasing to 10.17% after annealing. The

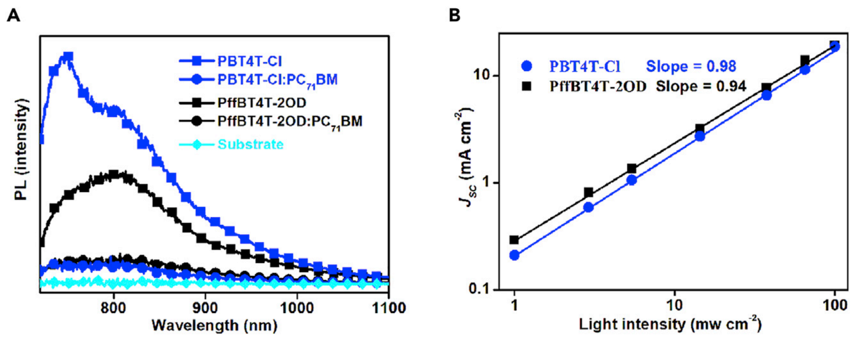


Figure 4. The Characteristics of Charge Dynamic in Polymer-Based Devices

(A) PL spectra of **PBT4T-Cl**, PffBT4T-OD, **PBT4T-Cl**:$PC_{71}BM$, and PffBT4T-OD:$PC_{71}BM$ films.

(B) J_{sc} versus light intensity (I) of the best PSC devices based on **PBT4T-Cl** and PffBT-2OD blended with $PC_{71}BM$.

pristine **PBT4T-Cl**:$PC_{71}BM$ blend film showed an RMS value of 15.98 nm (Figure 3C), with an optimized PCE of 11.18%, and the annealed film showed a much rougher surface, with an RMS of 24.43 nm. The excessive roughness with poor electrical conductivity because of the inferior interfacial contact causes unsatisfactory device performance. Thus, the chlorine substitution creates a driving force for the formation of favorable domain size of the active layer and intermolecular order stacking at the nanoscale, and thus it represents a feasible way to optimize the film morphology of blends used in PSCs.

As chlorine substitution in the polymer can cause enhanced intermolecular interactions and majority face-on orientation in the blend film, this suggested that additive-free PSCs might also present a high PCE. **PBT4T-Cl**-based PSCs with different ratios of DIO additive in chlorobenzene solvent were fabricated and their *J-V* curves and extracted device performance parameters are shown in Figure S12 and Table S5, respectively. As desired, the device without DIO exhibited a high PCE of 10.18%, which reveals the chlorinated **PBT4T-Cl** to be an excellent candidate for additive-free highly efficient PSCs. Compared with PffBT4T-2OD-based PSCs (Figure S13 and Table S6), the **PBT4T-Cl**-based PSCs also reduced the required DIO content in the processing solvent from 3% to 2% in the optimized devices. To reduce environmental hazards, a large number of non-halogenated, solvent-processed, high-efficiency PSCs have been reported in recent publications.[53–55] To promote the merits of chlorinated polymer-based PSCs, **PBT4T-Cl**-based PSCs were fabricated using non-halogenated solvents, including p-xylene and toluene. However, compared with the devices processed from mixed chlorobenzene (CB) and dichlorobenzene (DCB), the devices from non-halogenated solvents are still lagging. The PCEs of devices from p-xylene and toluene both exceed 9.35% (Figure S14 and Table S7), which is one of the highest reported efficiencies of chlorinated polymer donors fabricated with non-halogenated solvents.

Investigation of Charge Dynamics

To consider the exciton quenching effects in the chlorinated polymer films, the photoluminescence (PL) spectra were measured. Figure 4A shows the PL spectra of **PBT4T-Cl**, PffBT4T-2OD, and their blend films with $PC_{71}BM$. Based on the absorption profiles of the polymers (Figure S3), excitation wavelengths of 696 nm and 692 nm were selected for PffBT4T-2OD and **PBT4T-Cl**, respectively. The large and strong PL emission of PffBT4T-2OD films at approximately 800 nm is shown in Figure 4A, while the blend films had dramatically quenched PL spectra. **PBT4T-Cl** showed a similar trend, but the emission peak at 750 nm was clearly blue shifted

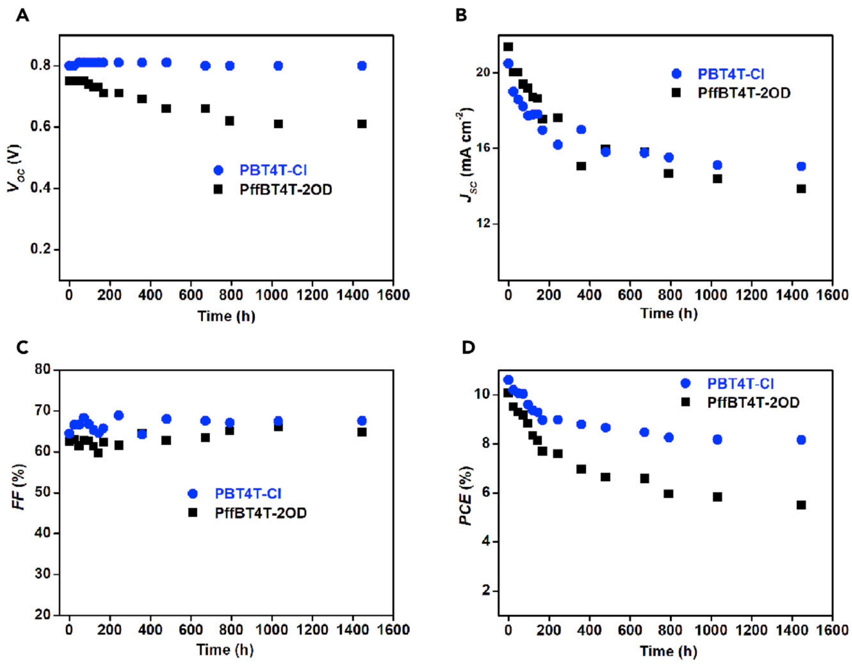


Figure 5. The Lifetime Tests of Optimal PSCs

(A–D) Performance parameters of unencapsulated devices stored in glove box.

(A) Open-circuit voltage (V_{OC}); (B) short-circuit current density (J_{sc}); (C) fill factor (FF); (D) power conversion efficiency (PCE).

and had greater quenching efficiency compared with PffBT4T-2OD, indicating the more-efficient charge transfer process with suppressive geminate recombination in the **PBT4T-Cl** blend films.[55] This evidence supports the higher FF of **PBT4T-Cl**-based PSCs as discussed above.

To further shed light on the effects of chlorination in conjugated polymers in terms of their charge recombination kinetics, we tested the light-intensity-dependent J_{sc} by manipulating the illumination intensity. As presented in Figure 4B, the relationship of J_{sc} to the light intensity (I) follows $J_{sc} \propto I^{\alpha}$.[56] The fitted α values for the control PffBT4T-2OD- and chlorinated **PBT4T-Cl**-based devices are 0.94 and 0.98, respectively. The high α indicates an efficient sweep-out of carriers occurs prior to recombination, in addition to an accelerated charge extraction observed from transient photocurrent measurement (Figure S15). The **PBT4T-Cl** devices showed a higher α value than the PffBT4T-2OD devices did, which indicated that **PBT4T-Cl** could reduce the bimolecular recombination and promote the exciton dissociation or charge collection in devices.[56] The dynamics views are in good agreement with the preferable morphology for the higher FF as described above, such that the chlorination of polymer is a viable path toward high-performance PSCs.

Device Stability

Currently, the PCEs of state-of-the-art PSCs have been sufficient for local commercialization, while greater stability is still urgently required for practical applications.[57,58] Developing more photochemically stable active materials would be one of many advances that would strengthen the stability of PSCs. Chlorination can enhance the stability of OFET devices, as previously reported.[40] To our knowledge, no report concerns the stability of chlorinated polymer-based PSCs. Therefore, we compared the device stabilities of the PffBT4T-2OD- and chlorinated

PBT4T-Cl-based devices, with both devices stored in a closed glove box with nitrogen atmosphere.

As shown in Figure 5, the basic parameters (including V_{OC}, J_{sc}, FF, and PCE) of the devices were plotted versus time. The V_{OC} of the **PBT4T-Cl**-based devices were highly stable at 0.80 V with almost no change during the course of the lifetime test. However, the V_{OC} of its non-chlorine analog dropped from 0.75 V to 0.61 V in the parallel experiment. The stable V_{OC} illustrated the enhanced stability of polymer via substituting a chlorine atom for an attached thiophene moiety. The drop in J_{sc} and FF arose from a range of mechanisms, including degradation of active layer, film morphological changes, and interlayer-electrode diffusion.[57] Considering all the situations, the **PBT4T-Cl**-based devices retained a PCE of 8.16% over 50 days, whereas the PffBT4T-2OD-based devices notably declined from 10.07% to 5.36%. From the plot of normalized PCE versus time (Figure S16), the **PBT4T-Cl**-based devices had better stability. The stability of devices without encapsulation presented a similar trend under ambient conditions (see Figures S17 and S18). To understand the reason for degradation, the time-dependent absorption spectra of blending films were conducted (Figures S19 and S20). The absorption profile of PffBT4T-2OD was much changed after 48 hr in the atmosphere. The more stable profile of **PBT4T-Cl** indicated that the chlorinated active layer possessed a steady two-phase nanostructure, resulting in enhanced stability of the devices. The improved stability of PSCs based on chlorinated polymers indicates that the chlorination of polymers has an important role in stabilizing the performance.

Conclusions

In summary, a novel chlorinated polymer donor **PBT4T-Cl** was designed and synthesized, and the PSCs using **PBT4T-Cl** showed a high PCE of 11.18%, which is the highest PCE of chlorinated polymer-based PSCs and is also one of the highest PCEs of FBT-based PSCs reported to date. The chlorination of polymers in PSCs will lead to enhancements both in V_{OC} and FF, resulting in further improvement of PCE, attributed to the fine-tuning of the molecular energy levels, morphology, and charge dynamics in devices by introducing chlorine atoms into suitable polymer backbones. Moreover, the chlorinated polymer-based PSCs exhibited favorable stability in the lifetime test compared with non-chlorinated analogs. Therefore, chlorination of the polymer donor is a feasible strategy to simultaneously increase the performance and stability of PSCs, eventually promoting the commercialization of PSCs.

EXPERIMENTAL PROCEDURES

Full experimental procedures are provided in the Supplemental Information.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, 25 figures, 7 tables, and 1 scheme and can be found with this article online at https://doi.org/10.1016/j.joule.2018.05.010.

ACKNOWLEDGEMENTS

The authors acknowledge financial support from SUSTech, the Recruitment Program of Global Youth Experts of China, the National Natural Science Foundation of China (51773087, 21733005), the Natural Science Foundation of Guangdong Province (2016A030313637), Shenzhen Fundamental Research program (JCYJ20170817111214740), and Shenzhen Nobel Prize Scientists Laboratory Project

(C17213101). W.C. gratefully acknowledges financial support from the US Department of Energy, Office of Science, Materials Science and Engineering Division.

AUTHOR CONTRIBUTIONS

F.H. and Z.H. designed and synthesized the polymers. H.C. fabricated and optimized the PSC devices. H.W. conducted the AFM and TEM. L.L. conducted the DFT calculations. P.C., J.Q., and A.L. contributed to material characterization. W.C. characterized GIWAXS. H.C. and F.H. integrated the interpretation and drafted the paper. F.H. conceived and directed the project. All authors commented on the final paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: March 1, 2018
Revised: April 30, 2018
Accepted: May 22, 2018
Published: June 13, 2018

REFERENCES

1. He, Z., Xiao, B., Liu, F., Wu, H., Yang, Y., Xiao, S., Wang, C., Russell, T.P., and Cao, Y. (2015). Single-junction polymer solar cells with high efficiency and photovoltage. Nat. Photonics *9*, 174–179.

2. Marrocchi, A., Facchetti, A., Lanari, D., Petrucci, C., and Vaccaro, L. (2016). Current methodologies for a sustainable approach to π-conjugated organic semiconductors. Energy Environ. Sci. *9*, 763–786.

3. Park, S.H., Roy, A., Beaupre, S., Cho, S., Coates, N., Moon, J.S., Moses, D., Leclerc, M., Lee, K., and Heeger, A.J. (2009). Bulk heterojunction solar cells with internal quantum efficiency approaching 100%. Nat. Photonics *3*, 297–302.

4. Li, Y. (2012). Molecular design of photovoltaic materials for polymer solar cells: toward suitable electronic energy levels and broad absorption. Acc. Chem. Res. *45*, 723–733.

5. Liu, Y., Zhao, J., Li, Z., Mu, C., Ma, W., Hu, H., Jiang, K., Lin, H., Ade, H., and Yan, H. (2014). Aggregation and morphology control enables multiple cases of high-efficiency polymer solar cells. Nat. Commun. *5*, 5293.

6. Zhang, Q., Kan, B., Liu, F., Long, G., Wan, X., Chen, X., Zuo, Y., Ni, W., Zhang, H., Li, M., et al. (2014). Small-molecule solar cells with efficiency over 9%. Nat. Photonics *9*, 35–40.

7. Sun, K., Xiao, Z., Lu, S., Zajaczkowski, W., Pisula, W., Hanssen, E., White, J.M., Williamson, R.M., Subbiah, J., Ouyang, J., et al. (2015). A molecular nematic liquid crystalline material for high performance organic photovoltaics. Nat. Commun. *6*, 6013.

8. Kan, B., Li, M., Zhang, Q., Liu, F., Wan, X., Wang, Y., Ni, W., Long, G., Yang, X., Feng, H., et al. (2015). A series of simple oligomer-like small molecules based on oligothiophenes for solution-processed solar cells with high efficiency. J. Am. Chem. Soc. *137*, 3886–3893.

9. Li, S., Ye, L., Zhao, W., Zhang, S., Mukherjee, S., Ade, H., and Hou, J. (2016). Energy-level modulation of small-molecule electron acceptors to achieve over 12% efficiency in polymer solar cells. Adv. Mater. *28*, 9423–9429.

10. Zhao, J., Li, Y., Yang, G., Jiang, K., Lin, H., Ade, H., Ma, W., and Yan, H. (2016). Efficient organic solar cells processed from hydrocarbon solvents. Nat. Energy *1*, 15027.

11. Yusoff, A.R.b. M., Kim, D., Kim, H.P., Shneider, F.K., da Silva, W.J., and Jang, J.A. (2015). High efficiency solution processed polymer inverted triple-junction solar cell exhibiting a power conversion efficiency of 11.83%. Energy Environ. Sci. *8*, 303–316.

12. Li, H., Xiao, Z., Ding, L., and Wang, J. (2018). Thermostable single-junction organic solar cells with a power conversion efficiency of 14.62%. Sci. Bull. *63*, 340–342.

13. Zhang, S., Qin, Y., Zhu, J., and Hou, J. (2018). Over 14% efficiency in polymer solar cells enabled by a chlorinated polymer donor. Adv. Mater. *30*, 1800868.

14. Xiao, Z., Jia, X., and Ding, L. (2017). Ternary organic solar cells offer 14% power conversion efficiency. Sci. Bull. *62*, 1562–1564.

15. Che, X., Li, Y., Qu, Y., and Forrest, S.R. (2018). High fabrication yield organic tandem photovoltaics combining vacuum-and solution-processed subcells with 15% efficiency. Nat. Energy *3*, 422–427.

16. Osaka, I., Shimawaki, M., Mori, H., Doi, I., Miyazaki, E., Koganezawa, T., and Takimiya, K. (2012). Synthesis, characterization, and transistor and solar cell applications of a naphthobisthiadiazole-based semiconducting polymer. J. Am. Chem. Soc. *134*, 3498–3507.

17. Guo, X., Zhou, N., Lou, S.J., Smith, J., Tice, D.B., Hennek, J.W., Ortiz, R.P., Navarrete, J.T.L., Li, S., Strzalka, J., et al. (2013). Polymer solar cells with enhanced fill factors. Nat. Photonics *7*, 825–833.

18. Huo, L., Zhang, S., Guo, X., Xu, F., Li, Y., and Hou, J. (2011). Replacing alkoxy groups with alkylthienyl groups: a feasible approach to improve the properties of photovoltaic polymers. Angew. Chem. Int. Ed. *50*, 9871–9876.

19. Zhang, M., Guo, X., Zhang, S., and Hou, J. (2014). Synergistic effect of fluorination on molecular energy level modulation in highly efficient photovoltaic polymers. Adv. Mater. *26*, 1118–1123.

20. Zhou, H., Yang, L., and You, W. (2012). Rational design of high performance conjugated polymers for organic solar cells. Macromolecules *45*, 607–632.

21. Elumalai, N.K., and Uddin, A. (2016). Open circuit voltage of organic solar cells: an in-depth review. Energy Environ. Sci. *9*, 391–410.

22. Xu, Z., Chen, L.-M., Yang, G., Huang, C.-H., Hou, J., Wu, Y., Li, G., Hsu, C.-S., and Yang, Y. (2009). Vertical phase separation in poly(3-hexylthiophene): fullerene derivative blends and its advantage for inverted structure solar cells. Adv. Funct. Mater. *19*, 1227–1234.

23. Nilsson, S., Bernasik, A., Budkowski, A., and Moons, E. (2007). .Morphology and phase segregation of spin-casted films of polyfluorene/PCBM blends. Macromolecules *40*, 8291–8301.

24. Matsumoto, F., Iwai, T., Moriwaki, K., Takao, Y., Ito, T., Mizuno, T., and Ohno, T. (2016). Controlling the polarity of fullerene derivatives to optimize nanomorphology in blend films. ACS Appl. Mater. Interfaces *8*, 4803–4810.

25. Ma, W., Tumbleston, J.R., Wang, M., Gann, E., Huang, F., and Ade, H. (2013). Domain purity and molecular orientation at donor/acceptor interfaces in high performance organic solar

cells: paths to further improvement. Adv. Energy Mater. *3*, 864–872.

26. Tumbleston, J.R., Collins, B.A., Yang, L., Stuart, A.C., Gann, E., Ma, W., You, W., and Ade, H. (2014). The influence of molecular orientation on organic bulk heterojunction solar cells. Nat. Photonics *8*, 385–391.

27. Chen, Z., Cai, P., Chen, J., Liu, X., Zhang, L., Lan, L., Peng, J., Ma, Y., and Cao, Y. (2014). Low band-gap conjugated polymers with strong interchain aggregation and very high hole mobility towards highly efficient thick-film polymer solar cells. Adv. Mater. *26*, 2586–2591.

28. Nian, L., Chen, Z., Herbst, S., Li, Q., Yu, C., Jiang, X., Dong, H., Li, F., Liu, L., Wurthner, F., et al. (2016). Aqueous solution processed photoconductive cathode interlayer for high performance polymer solar cells with thick interlayer and thick active layer. Adv. Mater. *28*, 7521–7526.

29. Zhang, Q., Kelly, M.A., Bauer, N., and You, W. (2017). The curious case of fluorination of conjugated polymers for solar cells. Acc. Chem. Res. *50*, 2401–2409.

30. Jo, J.W., Bae, S., Liu, F., Russell, T.P., and Jo, W.H. (2015). Comparison of two D−A type polymers with each being fluorinated on D and A unit for high performance solar cells. Adv. Funct. Mater. *25*, 120–125.

31. Wadsworth, A., Ashraf, R.S., Abdelsamie, M., Pont, S., Little, M., Moser, M., Hamid, Z., Neophytou, M., Zhang, W., Amassian, A., et al. (2017). Highly efficient and reproducible nonfullerene solar cells from hydrocarbon solvents. ACS Energy Lett. *2*, 1494–1500.

32. Hu, H., Jiang, K., Yang, G., Liu, J., Li, Z., Lin, H., Liu, Y., Zhao, J., Zhang, J., Huang, F., et al. (2015). Terthiophene-based D–A polymer with an asymmetric arrangement of alkyl chains that enables efficient polymer solar cells. J. Am. Chem. Soc. *137*, 14149–14157.

33. Wang, C., Xu, X., Zhang, W., Bergqvist, J., Xia, Y., Meng, X., Bini, Kim., Ma, W., Yartsev, K., Vandewal, K., et al. (2016). Low band gap polymer solar cells with minimal voltage losses. Adv. Energy Mater. *6*, 1600148.

34. Liu, J., Chen, S., Qian, D., Gautam, B., Yang, G., Zhao, J., Bergqvist, J., Zhang, F., Ma, W., Ade, H., et al. (2016). .Fast charge separation in a non-fullerene organic solar cell with a small driving force. Nat. Energy *1*, 16089.

35. Stuart, A.C., Tumbleston, J.R., Zhou, H., Li, W., Liu, S., Ade, H., and You, W. (2013). Fluorine substituents reduce charge recombination and drive structure and morphology development in polymer solar cells. J. Am. Chem. Soc. *135*, 1806–1815.

36. Deng, D., Zhang, Y., Zhang, J., Wang, Z., Zhu, L., Fang, J., Xia, B., Wang, Z., Lu, K., Ma, W., and Wei, Z. (2016). Fluorination-enabled optimal morphology leads to over 11% efficiency for inverted small-molecule organic solar cells. Nat. Commun. *7*, 13740.

37. Kawashima, K., Fukuhara, T., Suda, Y., Suzuki, Y., Koganezawa, T., Yoshida, H., Ohkita, H., Osaka, I., and Takimiya, K. (2016). Implication of fluorine atom on electronic properties, ordering structures, and photovoltaic performance in naphthobisthiadiazole-based semiconducting polymers. J. Am. Chem. Soc. *138*, 10265–10275.

38. Ryu, G.S., Chen, Z., Usta, H., Nor, Y.Y., and Facchetti, A. (2016). Naphthalene diimide-based polymeric semiconductors. effect of chlorine incorporation and n-channel transistors operating in water. MRS Commun. *6*, 47–60.

39. Facchetti, A. (2010). π-Conjugated polymers for organic electronics and photovoltaic cell applications. Chem. Mater. *23*, 733–758.

40. Tang, M.L., Oh, J.H., Reichardt, A.D., and Bao, Z. (2009). Chlorination: a general route toward electron transport in organic semiconductors. J. Am. Chem. Soc. *131*, 3733–3740.

41. Fleetham, T.B., Bakkan, N., Mudrick, J.P., Myers, J.D., Cassidy, V.D., Cui, J., Xue, J., and Li, J. (2013). Enhanced open-circuit voltage in organic photovoltaic cells with partially chlorinated zinc phthalocyanine. J. Mater. Sci. *48*, 7104–7114.

42. Zheng, Y.-Q., Wang, Z., Dou, J.-H., Zhang, S.-D., Luo, X.-Y., Yao, Z.-F., Wang, J.-Y., and Pei, J. (2015). Effect of halogenation in isoindigo-based polymers on the phase separation and molecular orientation of bulk heterojunction solar cells. Macromolecules *48*, 5570–5577.

43. Li, Y., Lin, J.D., Che, X., Qu, Y., Liu, F., Liao, L.S., and Forrest, S.R. (2017). High efficiency near-infrared and semitransparent non-fullerene acceptor organic photovoltaic cells. J. Am. Chem. Soc. *139*, 17114–17119.

44. Guo, B., Li, B., Guo, X., Meng, X., Ma, W., Zhang, M., and Li, Y. (2017). High efficiency nonfullerene polymer solar cells with thick active layer and large area. Adv. Mater. *29*, 1702291.

45. Carsten, B., Szarko, J.M., Son, H.J., Wang, W., Lu, L., He, F., Rolczynski, B.S., Lou, S., Chen, L., and Yu, L. (2011). Examining the effect of the dipole moment on charge separation in donor-acceptor polymers for organic photovoltaic applications. J. Am. Chem. Soc. *133*, 20468–20475.

46. Cho, H.H., Kim, S., Kim, T., Sree, V.G., Jin, S.H., Kim, F.S., and Kim, B.J. (2017). Design of cyanovinylene-containing polymer acceptors with large dipole moment change for efficient charge generation in high-performance all-polymer solar cells. Adv. Energy Mater. *7*, 1701436.

47. Lu, L., and Yu, L. (2014). Understanding low bandgap polymer PTB7 and optimizing polymer solar cells based on it. Adv. Mater. *26*, 4413–4430.

48. Kranthiraja, K., Kim, S., Lee, C., Gunasekar, K., Sree, V.G., Gautam, B., Gundogdu, K., Jin, S.H., and Kim, B.J. (2017). Comparison of two D−A type polymers with each being fluorinated on D and A unit for high performance solar cells. Adv. Funct. Mater. *27*, 1701256.

49. Murgatroyd, P.N. (1970). Theory of space-charge-limited current enhanced by Frenkel effect. J. Phys. D Appl. Phys. *3*, 151–156.

50. He, Z., Zhong, C., Huang, X., Wong, W.Y., Wu, H., Chen, L., Su, S., and Cao, Y. (2011). Simultaneous enhancement of open-circuit voltage, short-circuit current density, and fill factor in polymer solar cells. Adv. Mater. *23*, 4636–4643.

51. Chen, W., Xu, T., He, F., Wang, W., Wang, C., Strzalka, J., Liu, Y., Wen, J., Miller, D.J., Chen, J., et al. (2011). Hierarchical nanomorphologies promote exciton dissociation in polymer/fullerene bulk heterojunction solar cells. Nano Lett. *11*, 3707–3713.

52. Hu, Z., Chen, H., Qu, J., Zhong, X., Chao, P., Xie, M., Lu, W., Liu, A., Tian, L., Su, Y., et al. (2017). Design and synthesis of chlorinated benzothiadiazole-based polymers for efficient solar energy conversion. ACS Energy Lett. *2*, 753–758.

53. Zhou, Y., Gu, K., Gu, X., Kurosawa, T., Yan, H., Guo, Y., Koleilat, G.I., Zhao, D., Toney, M.F., and Bao, Z. (2016). All-polymer solar cells employing non-halogenated solvent and additive. Chem. Mater. *28*, 5037–5042.

54. Duan, C., Gao, K., Van Franeker, J.J., Liu, F., Wienk, M.M., and Janssen, R.A. (2016). Toward practical useful polymers for highly efficient solar cells via a random copolymer approach. J. Am. Chem. Soc. *138*, 10782–10785.

55. Yuan, J., McDowell, C., Mai, C.-K., Bazan, G.C., and Ma, W. (2016). Ternary D1–D2–A–D2 structured conjugated polymer: efficient "green" solvent-processed polymer/neat-C_{70} solar cells. Chem. Mater. *28*, 7479–7486.

56. Kyaw, A.K.K., Wang, D., Luo, C., Cao, Y., Nguyen, T.-Q., Bazan, G.C., and Heeger, A.J. (2014). Effects of solvent additives on morphology, charge generation, transport, and recombination in solution-processed small-molecule solar cells. Adv. Energy Mater. *4*, 1301469.

57. Park, K.H., An, Y., Jung, S., Park, H., and Yang, C. (2016). The use of an n-type macromolecular additive as a simple yet effective tool for improving and stabilizing the performance of organic solar cells. Energy Environ. Sci. *9*, 3464–3471.

58. You, J., Meng, L., Song, T.B., Guo, T.F., Yang, Y.M., Chang, W.H., Hong, Z., Chen, H., Zhou, H., Chen, Q., et al. (2016). Improved air stability of perovskite solar cells via solution-processed metal oxide transport layers. Nat. Nanotechnol. *11*, 75–81.